

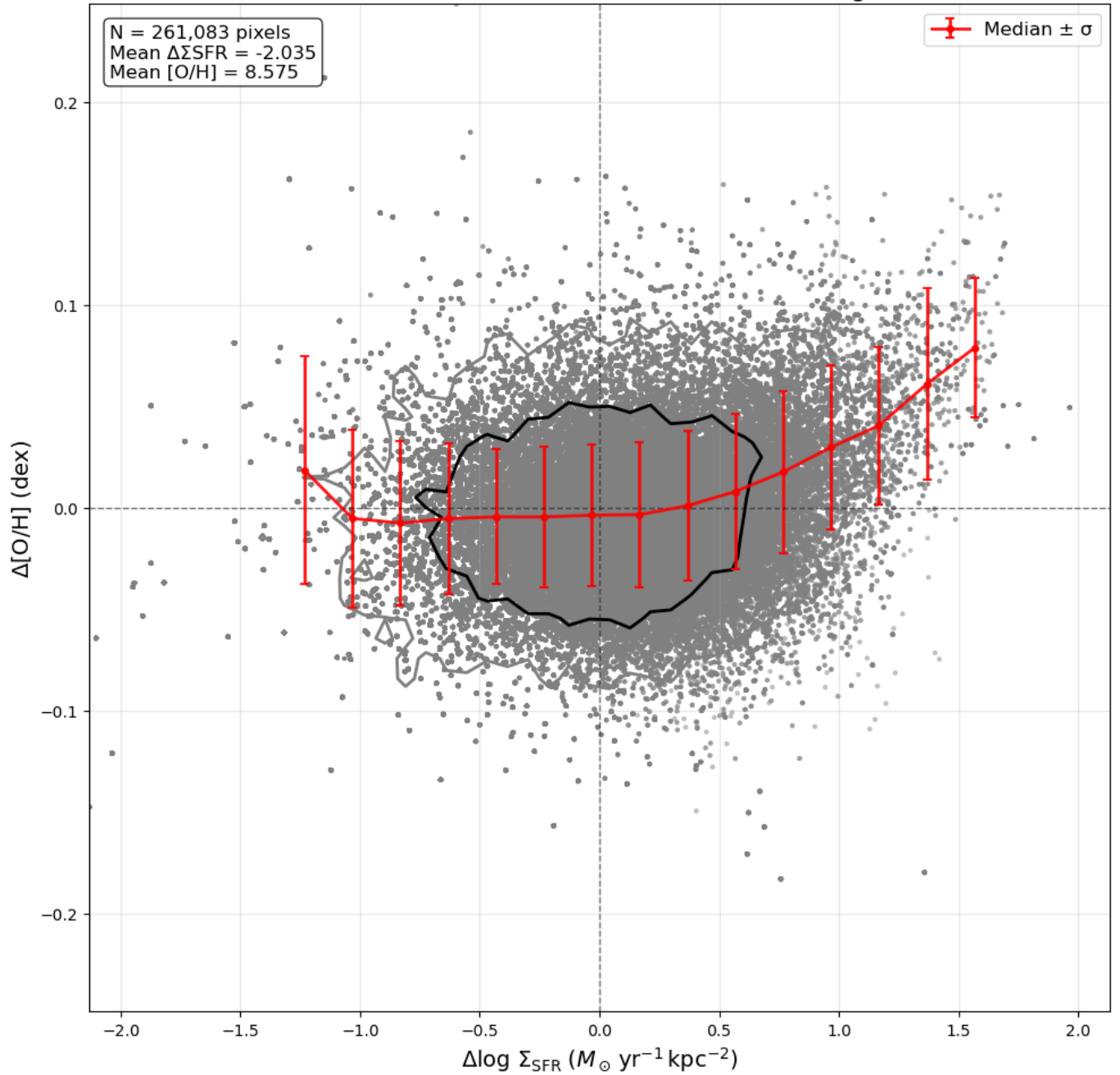
20251016 Slope Map from Bin Average in HII

Offset relation: $\Delta[\text{O}/\text{H}]$ vs $\Delta\Sigma_{\text{SFR}}$

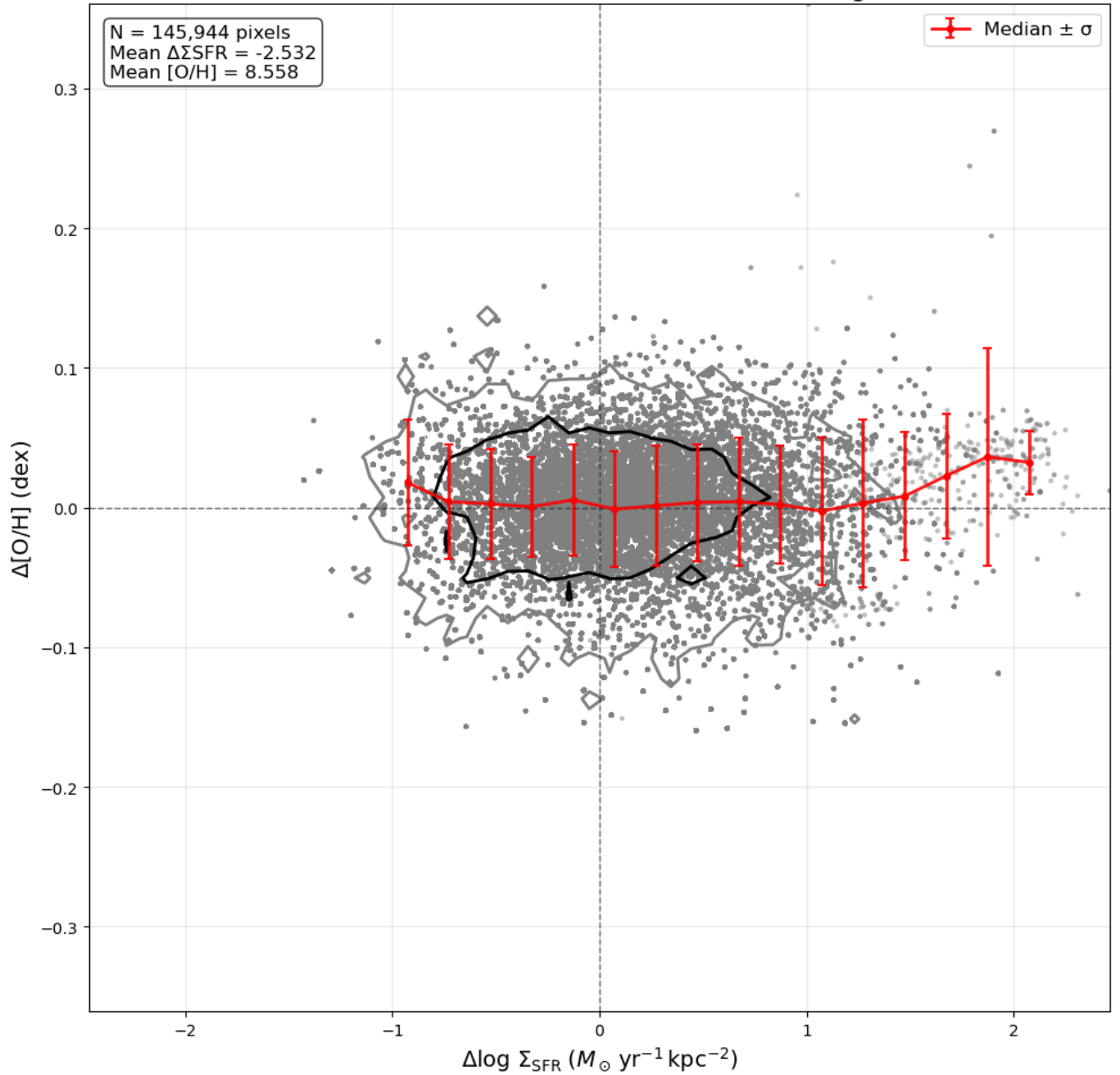
Note that the offset here is defined as the offset from the average of each Σ_* bin. We first divide data into 12 equal-width Σ_* bin. Then in each Σ_* bin, we subtract Σ_{SFR} and $[\text{O}/\text{H}]$ by the mean value to obtain the offset, $\Delta\Sigma_{\text{SFR}}$ and $\Delta[\text{O}/\text{H}]$.

Below I show the $\Delta[\text{O}/\text{H}]$ vs $\Delta\Sigma_{\text{SFR}}$ relations for NGC4501, NGC4192, NGC4330 and NGC4396.

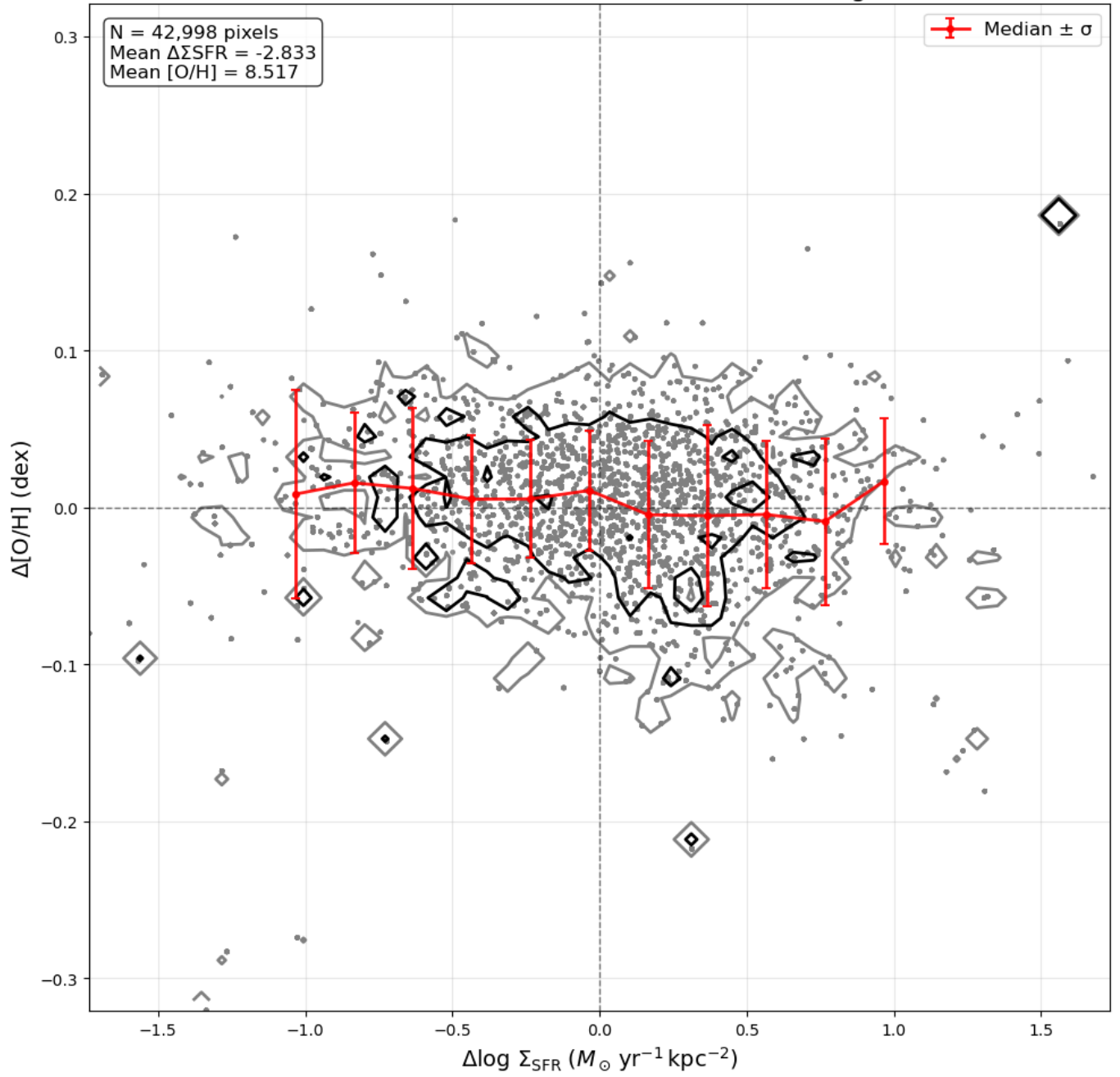
NGC4501: $\Delta[\text{O}/\text{H}]$ vs $\Delta\Sigma\text{SFR}$ (O3N2-M13, HII regions)

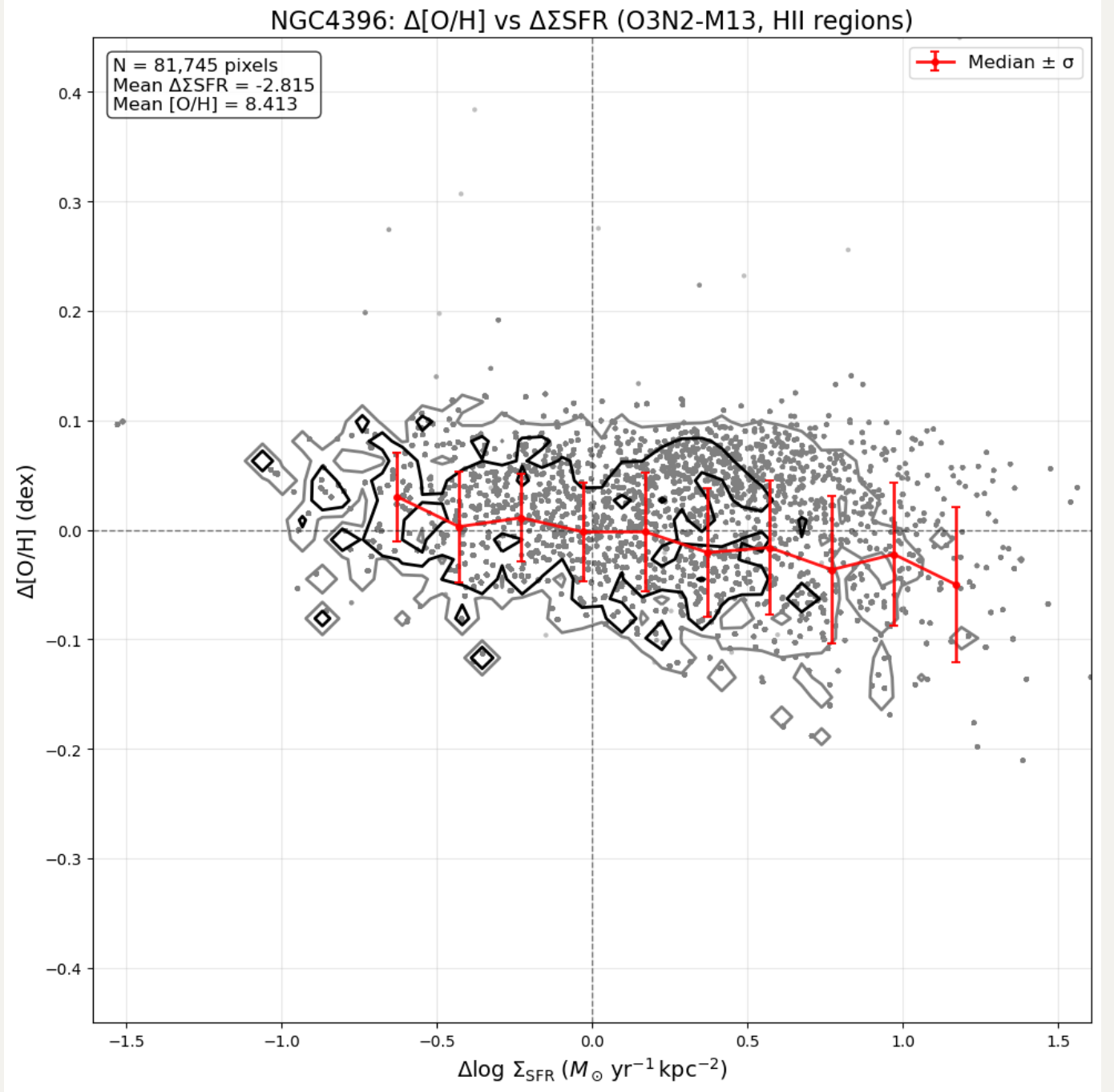


NGC4192: $\Delta[\text{O}/\text{H}]$ vs $\Delta\Sigma_{\text{SFR}}$ (O3N2-M13, HII regions)



NGC4330: $\Delta[\text{O}/\text{H}]$ vs $\Delta\Sigma\text{SFR}$ (O3N2-M13, HII regions)





Slope or Ratio map

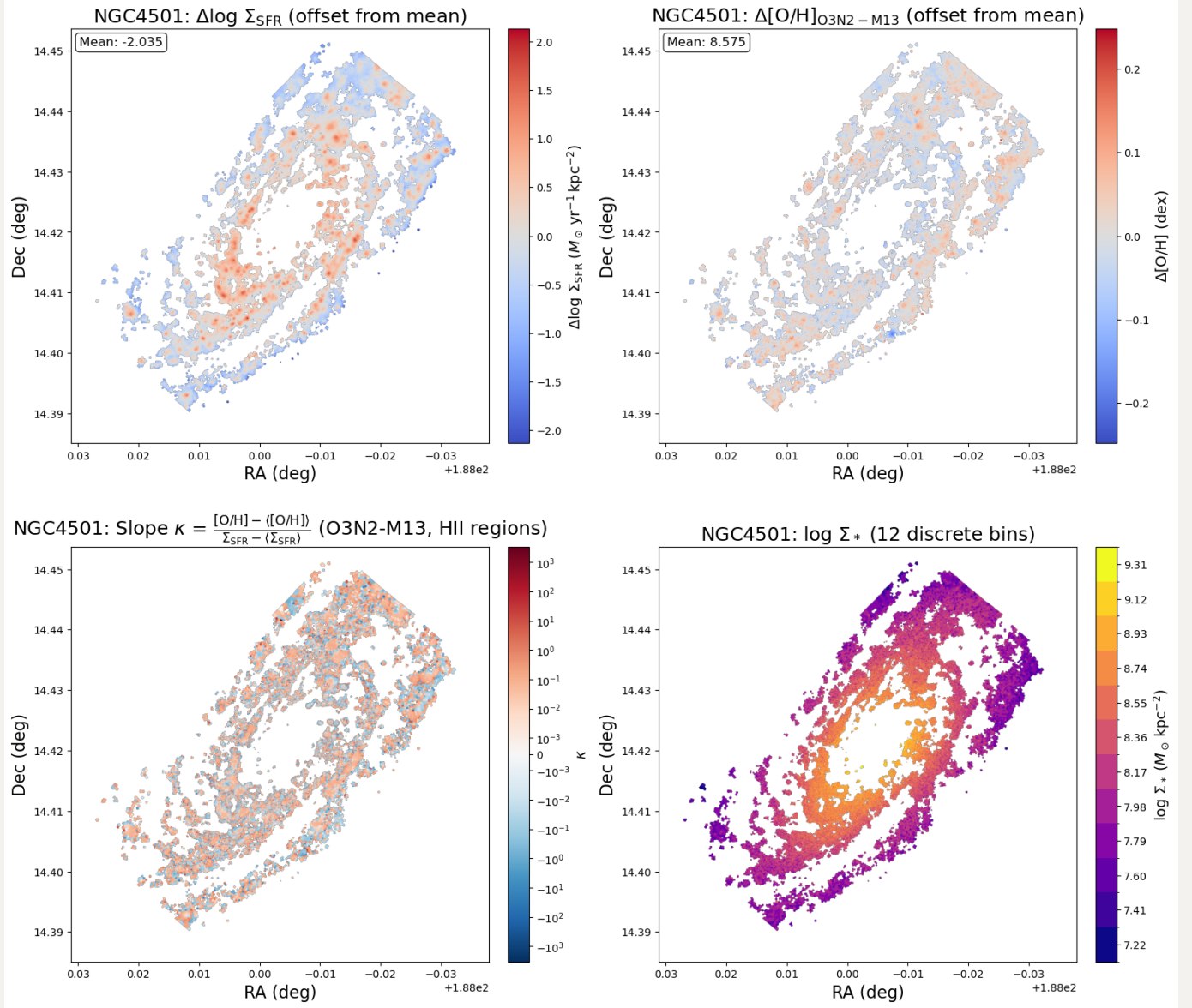
To further identify the correlation between $[\text{O}/\text{H}]$ vs Σ_{SFR} after remove the Σ_* dependence (or $\Delta[\text{O}/\text{H}]$ vs $\Delta\Sigma_{\text{SFR}}$), I define the slope or ratio from $[\text{O}/\text{H}]$ vs Σ_{SFR} (or $\Delta[\text{O}/\text{H}]$ vs $\Delta\Sigma_{\text{SFR}}$) relation of each Σ_* bin:

$$\kappa = \frac{[\text{O}/\text{H}] - \langle [\text{O}/\text{H}] \rangle}{\Sigma_{\text{SFR}} - \langle \Sigma_{\text{SFR}} \rangle} = \frac{\Delta[\text{O}/\text{H}]}{\Delta\Sigma_{\text{SFR}}} \quad (1)$$

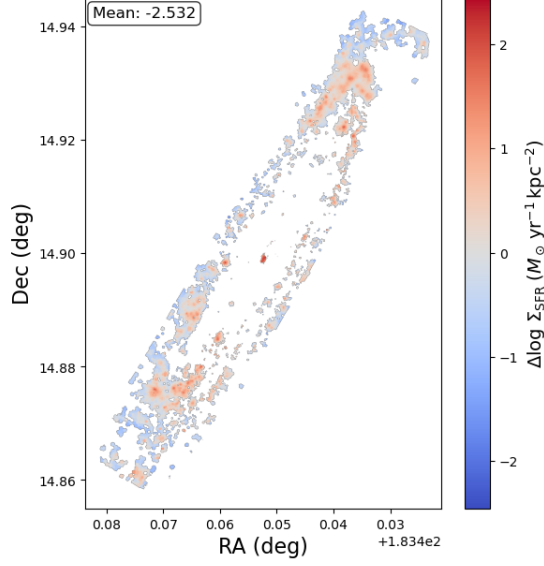
But taking the slope or ratio, we can now:

1. positive or negative correlation by the sign of κ
2. how positive or negative the correlation is by the magnitude of κ

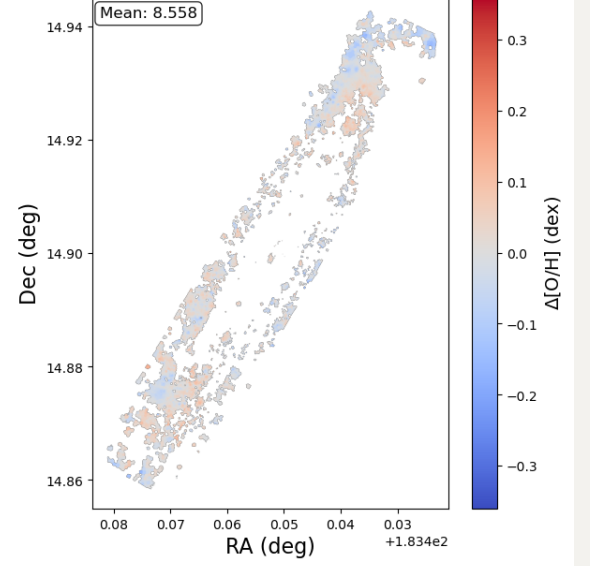
As usual, I show the κ map in the bottom left for NGC4501, NGC4192, NGC4330 and NGC4396.



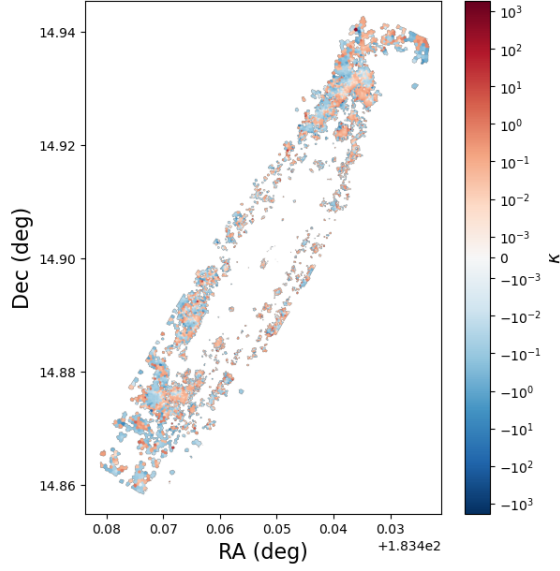
NGC4192: $\Delta \log \Sigma_{\text{SFR}}$ (offset from mean)



NGC4192: $\Delta[\text{O}/\text{H}]_{\text{O3N2-M13}}$ (offset from mean)



NGC4192: Slope $\kappa = \frac{[\text{O}/\text{H}] - \langle [\text{O}/\text{H}] \rangle}{\Sigma_{\text{SFR}} - \langle \Sigma_{\text{SFR}} \rangle}$ (O3N2-M13, HII regions)



NGC4192: $\log \Sigma_{*}$ (12 discrete bins)

